

Lecture 7: GROUPS!

Definition: A group $(G, *)$ is an algebraic structure i.e:

- G is a set,
- $*: G \times G \rightarrow G$ is a binary operation and we require that the following 3 axioms are satisfied:

• Axiom 1: the operation is associative.

$\forall a, b, c \in G$ we have

$$a * (b * c) = (a * b) * c.$$

• Axiom 2: existence of an identity element

~~$\forall a \in G$~~ $\exists e \in G$ s.t. $\forall a \in G$

$$a * e = e * a = a.$$

• Axiom 3: Existence of inverses.

Let e be the identity of G .

$\forall a \in G \exists b \in G$ such that

$$a * b = b * a = e.$$

SET	OPERATION	IDENTITY	INVERSES	Examples:
1) $\mathbb{Z}$ ($\mathbb{Z}, +, 0$)	+ sum	0	For z take $-z$. Ab	
2) $\mathbb{Q}^\times$ ($\mathbb{Q}^\times, \cdot, 1$)	• multiplication	1	For q take $\frac{1}{q}$. Ab	
3) $U_1 = \{z \in \mathbb{C} \mid z = 1\}$ ($U_1, \cdot, 1$)	• complex multiplication $e^{i\theta_1} \cdot e^{i\theta_2} = e^{i(\theta_1 + \theta_2)}$	$1 = e^{i \cdot 0}$	For $e^{i\theta}$ take $e^{i(2\pi - \theta)}$ Ab	
4) $U_n = \{z \in \mathbb{C} \mid z = 1 \text{ and } z^n = 1\}$ Recall we describe them $e^{i\frac{2\pi}{n} \cdot k}$ for $k = 0, \dots, n-1$	• complex multiplication $e^{i(\frac{2\pi}{n} \cdot \ell)} \cdot e^{i(\frac{2\pi}{n} \cdot k)} = e^{i(\frac{2\pi}{n}(\ell + k))}$	$1 = e^{i0}$	For $e^{i\frac{2\pi}{n} \cdot k}$ take $e^{i\frac{2\pi}{n}(n-k)}$ Ab	
5) $GL_n(\mathbb{R}) = \{A \text{ is an } n \times n \text{ matrix with } \det(A) \neq 0\}$	Matrix multiplication $GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$ $(A, B) \mapsto A \cdot B$	$I = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$	For A take A^{-1} Not Abelian!	

More examples:

SET	Group	Operation	Identity	Inverses
$\mathbb{R}/m\mathbb{Z} = \{[0], \dots, [m-1]\}$ $M_{n \times n}(\mathbb{R}) = \{A \text{ matrix on } n \times n \text{ entries in } \mathbb{R}\}$	$(\mathbb{R}/m\mathbb{Z}, +, [0])$ $(M_{n \times n}(\mathbb{R}), +, 0)$	$+: \mathbb{R}/m\mathbb{Z} \times \mathbb{R}/m\mathbb{Z} \rightarrow \mathbb{R}/m\mathbb{Z}$ $[a], [b] \mapsto [a+b]$ $+: M_{n \times n}(\mathbb{R}) \times M_{n \times n}(\mathbb{R}) \rightarrow M_{n \times n}(\mathbb{R})$ $(A, B) \mapsto A+B.$	$[0]$ 0-matrix $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	$[r]$ take $[m-r]$. $Ab.$ $Ab.$ For A take $-A.$

Another example: $\mathbb{Z}/m\mathbb{Z}$ where $m\mathbb{Z}$ is the congruence of divisibility by m .

$$\mathbb{Z}/m\mathbb{Z} = \{[0], \dots, [m-1]\}$$

with the operation: $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$
 $([a], [b]) \rightarrow [a+b].$

well defined:

- identity is $[0]$
- inverses: let $[r]$ for $0 \leq r \leq m-1$
have $[m-r]$.

Definition: A group G is abelian if its binary operation is commutative.

Add here Definition of homomorphisms of groups!

Properties of groups:

thm 1 Let G be a group $(G, *, e)$ then the left and right cancellation laws hold in G , that is

- $a * b = a * c$ implies $b = c$
- $b * a = c * a$ implies $b = c$.

Proof:

suppose $a * b = a * c$, then we multiply both sides by a^{-1}

$$a^{-1} * (a * b) = a^{-1} * (a * c)$$

} associativity.

$$(a^{-1} * a) * b = (a^{-1} * a) * c$$

} inverses

$$e * b = e * c$$

} e is the identity.

$$b = c.$$

the other case is similar.

thm 2: Let $(G, *, e)$ be a group if $a, b \in G$ then the equations:

- $a * x = b$ always have a solution
- $y * a = b$ x, y in G .

Proof: Let $a * x = b$ for $a, b \in G$
we want to find $x \in G$ that satisfies $*$.

How to find x ? Well start trying how to isolate it.

$$a * x = b \quad \Leftrightarrow \quad a^{-1} * (a * x) = a^{-1} * b$$

↑
multiply by a^{-1}

by associativity $(a^{-1} * a) * x = a^{-1} * b.$

by inverses $e * x = a^{-1} * b.$

by identity

$$x = a^{-1} * b.$$

Now copy the
proof upwards

For the second part the argument is similar

Thm 3: In a group $(G, *, e)$ with binary operation
there is only one element e such that

$$e * x = x * e = x \quad \forall x \in G.$$

Likewise for each $a \in G$ there is only one element
 $a' \in G$ such that $a' * a = a * a' = e.$

i.e. The identity element and inverses are
UNIQUE.

Proof:

• uniqueness of identity:

Assume you are given two identities e, e'

Then

$$e = e * e' = e'$$

$\uparrow$
 e' is an identity $\quad e$ is an identity.

• uniqueness of inverses: Let $a \in G$ and assume there are b_1 and b_2 s.t.

$$a * b_1 = b_1 * a = e \quad (1)$$

$$a * b_2 = b_2 * a = e.$$

Now From (1) $a * b_1 = b_1 * a = e$

$b_1 * a = e$ multiply b_2 on the right

$$(b_1 * a) * b_2 = b_2$$

"ass.

$$b_1 * (a * b_2) = b_2$$

$$b_1 * \underbrace{e}_{= e} = b_2 \Rightarrow b_1 = b_2.$$

Corollary: ~~let $a, b \in G$ then~~ let G be a group and $a, b \in G$ then $(a * b)^{-1} = b^{-1} * a^{-1}$.

Lecture 8: Subgroups, cyclic groups and generators.

Definition: Let G be a group $(G, *, e)$ and

$H \subseteq G$ such that:

1. $e \in H$, (has the identity)

2. $\forall x, y \in H \Rightarrow x * y \in H$ (closed under $*$)

3. $\forall x \in H \Rightarrow x^{-1} \in H$. (closed under inverses)

then H is said to be a subgroup of G .

Analogue: subspace of a vector space.

Remarks: 1. A subgroup is naturally equipped with a group structure i.e. $(H, *, e)$ is a group.

Examples: We have seen a few.

1) $(\mathbb{Z}, +)$ is a subgroup of $(\mathbb{R}, +, 0)$

2) If V is a vector space over $\mathbb{R}$ then $(V, +, \vec{0})$ is an abelian group. Let

$W \subseteq V$ be a subspace then $(W, +, \vec{0})$ is a subgroup of $(V, +, \vec{0})$

Rmk: I don't care about scalar multiplication.

3. $(U, \cdot, 1)$ and $(U_n, \cdot, 1)$ are subgroups of $(\mathbb{C}, \cdot, 1)$

4. Another example: consider $(GL_n(\mathbb{R}), \cdot, I)$
this is a group.

$$\text{Let } H = \{A \in GL_n(\mathbb{R}) \mid A^T A = A A^T = I\}$$

$\uparrow$
it is called the unitary group.

it is a subgroup of $(GL_n(\mathbb{R}), \cdot, I)$

Proof:

• closed under $\cdot$: let $A, B \in H$ we want to show $A \cdot B \in H$.

we have $\left. \begin{array}{l} (1) A \cdot A^T = A^T A = I \\ (2) B B^T = B^T B = I \end{array} \right\}$ by hypothesis.

Recall: $(AB)^T = B^T \cdot A^T$.

$$\text{Then } (A \cdot B) \cdot (A \cdot B)^T = (A \cdot B)(B^T A^T)$$

$$= A \cdot (B \cdot B^T) \cdot A^T = A \cdot I \cdot A^T = A A^T = I.$$

$\underbrace{\quad}_{I \text{ by (2)}}$

• identity: $I^T = I$.

• inverses:

closed under inverses: let $A \in H$ we want

to show $A^{-1} \in H$. since $A \in H$ then $A^{-1} = A^T$

we want to check $A^T \in H$.

This means $A^T \cdot (A^T)^T = (A^T)^T = A^T = I$.

Recall: $(A^T)^T = A \Rightarrow A^T \in H$.

5. ~~to~~ Fix $m \in \mathbb{N}$ and let $(m\mathbb{Z}, +, 0)$ this is a subgroup of $(\mathbb{Z}, +, 0)$

Proof: We must check 3 things:

1) identity is in the set: $0 \in m\mathbb{Z}$ since $0 = m \cdot 0$

2) closed under +: let $z_1, z_2 \in m\mathbb{Z}$ then they are $w_1, w_2 \in \mathbb{Z}$ s.t.

(1) $z_1 = m \cdot w_1$ and $z_2 = m \cdot w_2$. (2)

our goal: $z_1 + z_2 \in m\mathbb{Z}$ i.e. $\exists c$ $z_1 + z_2 = m \cdot c$.

(1) + (2) : $z_1 + z_2 = mw_1 + mw_2 = m(w_1 + w_2)$
 $\underbrace{w_1 + w_2}_{= c \in \mathbb{Z}}$

3) closed under inverse: -

let $z \in m\mathbb{Z}$ then there is $w \in \mathbb{Z}$ s.t.

$z = m \cdot w$ then $-z = m \cdot \underbrace{(-w)}_{\in \mathbb{Z}} \in m\mathbb{Z}$.

Definition:

Cyclic subgroups:

Theorem: Let $(G, *, e)$ be a group then

$H = \{a^n \mid n \in \mathbb{Z}\}$ is a subgroup of G

where a is a fixed element in G

Notation: • $a^n = \underbrace{a * \dots * a}_{n\text{-times.}}$

and many times
we just write it
with a dot.

• for $n \in \mathbb{Z}_{<0}$ $a^n = \underbrace{a^{-1} * \dots * a^{-1}}_{-n\text{-times.}}$

• additive na

Example: suppose we work in $\mathbb{Z}/8\mathbb{Z}$

~~$= \{0, 1, \dots, 7\}$~~

~~$H = \{0, 2, 2, 2, 6, 4\}$~~

~~For $a = 2$ $2+2=4$~~

~~$2^{-1} = 6$ since $2+6=8$.~~

~~$2^{-2} = 4^{-1} = 4$ since $4+4=8$~~

~~so $H = \{0, 2, 4, 6\}$~~

~~$6+6+6=18 \equiv_8 2$~~

~~$6+6+6+6=24 \equiv_8 0$~~

~~$\equiv_8 0$~~

Example: 1) $\mathbb{Z}/8\mathbb{Z} \cong \{0, 2, 4, 6\}$ is a subgroup.

$a=2$	0	$1 \cdot 2$	$2 \cdot 2$	$3 \cdot 2$	$4 \cdot 2$
-6	0	2	$2+2$	$2+2+2$	$4+2$
2	0	2	4	6	

Proof: In general.

2) take $(\mathbb{R}, \cdot, 1)$

$$H = \{\pi^n \mid n \in \mathbb{Z}\}$$

=

$$\dots \pi^{-1} \cdot \pi \cdot \pi^2 \cdot \pi^3 \dots$$

$$\begin{aligned} \bullet \text{ "2" } &= 6 \\ \hookrightarrow 2+6 &= 8 \end{aligned}$$

$$\begin{aligned} \bullet -4 &= 4 \\ \text{since} \\ 4+4 &= 8 \end{aligned}$$

$$\begin{aligned} \bullet \text{ "6" } &= 2 \\ \text{since } 2+6 &= 8 \end{aligned}$$

Proof of the theorem:

~~closed under~~.

• identity: $e = a^0$ for every $a \in G$ by convention

• closed under $\cdot$: Let $n, m \in \mathbb{Z}$ and $a^n, a^m \in H$ then $a^n \cdot a^m = a^{n+m}$

$$\underbrace{a \cdot \dots \cdot a}_{n \text{ times}} \cdot \underbrace{a \cdot \dots \cdot a}_{m \text{ times}} = \underbrace{a \cdot \dots \cdot a}_{n+m \text{ times}}$$

• closed under inverse: $n \in \mathbb{Z}$ and $a^n \in H$ then $a^{-n} \in H$ and $a^n \cdot a^{-n} = a^0 = e$.

In this case we say that H is the cyclic group generated by a , and write $H = \langle a \rangle$ and refer to a as the generator of H .

↓
this definition is inspired in $(\mathbb{Z}, +, 0)$ which is generated by 1. $\mathbb{Z} = \langle 1 \rangle$.

Definition: Let $(G, *, e)$ be a group we say that G is cyclic if there is $a \in G$ such that $G = \langle a \rangle = \{a^n \mid n \in \mathbb{Z}\}$, i.e. it is generated by a single element.

More examples on cyclic groups:

We have so far:

1) $(\mathbb{Z}, +, 0) = \langle 1 \rangle$

2) $\{0, 2, 4, 6\} = \langle 2 \rangle$ inside $(\mathbb{Z}/8\mathbb{Z}, +, 0)$

3) $\{\pi^n \mid n \in \mathbb{Z}\} = \langle \pi \rangle$ inside $(\mathbb{R}, \cdot, 1)$

4) $(m\mathbb{Z}, +, 0)$ inside $(\mathbb{Z}, +, 0)$ is generated by $\langle m \rangle$.

5) $(U_n, \cdot, 1)$ is generated by e
 $e = \frac{2\pi i}{n}$

obtained
by the
theorem.

Remark for myself: state the definition before and give 1, 2 and 3 as examples.

Then the theorem and produce 4

→ Lecture 9 The structure of cyclic groups.

Recall A group $(G, *, e)$ is called cyclic if it is the group generated by a single element a i.e. $G = \{a^n \mid n \in \mathbb{Z}\}$
(in additive notation
 $G = \{na \mid n \in \mathbb{Z}\}$)

Main theorem of past time: Let $(G, *, e)$ be a group. and $a \in G$ then

$H = \{a^n \mid n \in \mathbb{Z}\}$ is a subgroup.

i.e. $H = \langle a \rangle$ is always a subgroup

- H is said to be generated by a ,
- a is called the generator of H .

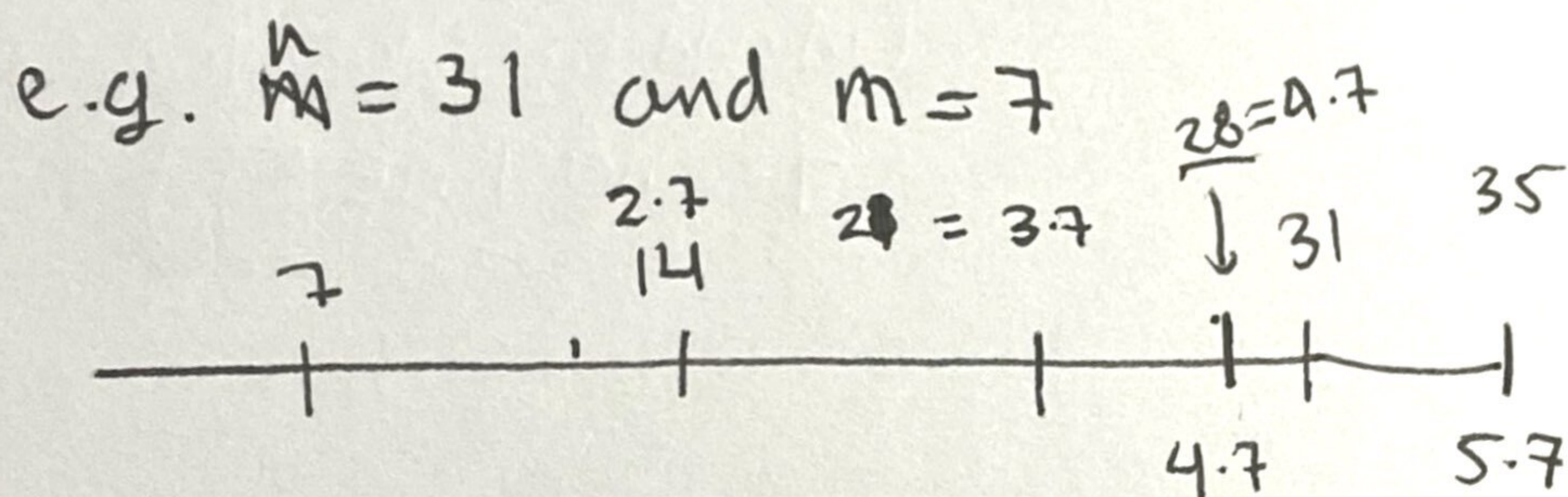
OUR GOAL: WHAT CAN BE SAID ABOUT THE STRUCTURE OF A CYCLIC GROUP?

Thm 1: Every cyclic group is abelian.

Proof: Let $g_1, g_2 \in G$ and assume $G = \langle a \rangle$
then they are $n, m \in \mathbb{Z}$ s.t. $g_1 = a^n$ and $g_2 = a^m$
then $g_1 \cdot g_2 = a^n \cdot a^m = a^{n+m} = a^{m+n} = a^m \cdot a^n = g_2 \cdot g_1$

the division algorithm: If $m \in \mathbb{Z}_{>0}$ and $n \in \mathbb{N}$ there are unique integers q and r such that

$$n = mq + r \quad \text{and} \quad 0 \leq r < m.$$



$$31 = 4 \cdot 7 + 3 \quad \begin{matrix} r=3 \\ q=4. \end{matrix}$$

Proof: let q be such that $mq \leq n < m(q+1)$
 Then $\exists r \quad 0 \leq r < m$ s.t. $n = mq + r$.

Notation and terminology:

- q is said to be the quotient of n modulo m
- r is said to be the remainder, of n modulo m .

Theorem: A subgroup of a cyclic group is cyclic.

Proof: $G = \langle a \rangle = \{a^n \mid n \in \mathbb{Z}\}$.

And let $H \leq G$ let $m \in \mathbb{Z}_{\geq 0}$ be the smallest positive integer s.t. $a^m \in H$.

Claim: Every other element in H is generated by a^m i.e. it is a power of a^m .

Let $n \in \mathbb{Z}$ be such that $a^n \in H$.

By the division algorithm $\exists! q, r$ $0 \leq r < m$ s.t. $n = qm + r$.

then $a^n = a^{qm+r} = a^{qm} \cdot a^r$

If we show $r=0$ we are done! since in this case $a^n = (a^m)^q$.

But why? ~~If not~~: $a^n \in H$ and $a^m \in H$

then $a^{qm} \in H \Rightarrow a^r = a^n \cdot (a^{qm})^{-1} \in H$

so $a^r \in H$ but we chose m minimal!

so $r=0$.

As required.

Corollary: the subgroups of $(\mathbb{Z}, +, 0)$ are the groups $(m\mathbb{Z}, +, 0)$ for $m \in \mathbb{N}$.

Recall: For $n, m \in \mathbb{Z}$.

- highest common divisor: n, m is the greatest positive integer that divides both n and m .
we write $\text{HCD}(n, m)$
- n, m are relatively prime if $\text{HCD}(n, m) = 1$.

FACT Given $n, m \in \mathbb{Z} \exists u, v \in \mathbb{Z}$ s.t.
 $\text{HCD}(n, m) = u \cdot n + v \cdot m$.

Definition Let $(G, *, e)$ be a group ~~that~~
~~is finite~~. then $|G| = \# \text{ of elements of } G = \text{the order of } G$.

CHARACTERIZATION OF CYCLIC GROUPS.

Thm 2 Let G be a cyclic group with generator a . If the order of G is infinite, then G is $(\mathbb{Z}, +, 0)$.

If the order of G is finite then there is an $n \in \mathbb{N}$ s.t. $G \cong (\mathbb{Z}_n, +_n, 0)$.

Proof: Let G be a cyclic group generated by a .

Claim 1: If G is ^{infinite} isomorphic to $(\mathbb{Z}, +, 0)$

Remark: If G is cyclic then

G is infinite $\iff a^m \neq e \ \forall m \in \mathbb{Z}$.

Thus if G is infinite:

$f: \mathbb{Z} \longrightarrow G$ is an isomorphism.
 $z \longmapsto a^z$

• injective: if $a^{z_1} = a^{z_2} \iff a^{z_1 - z_2} = a^0 = e$

$\implies a^{z_1 - z_2} = e$

$\uparrow$

RMK

• surjective: by def.

• homomorphism: $f(z_1 + z_2) = a^{z_1 + z_2} = a^{z_1} \cdot a^{z_2}$
 $= f(z_1) \cdot f(z_2)$.

Case 2: G is finite.

Let m_0 be the smallest positive integer s.t.

$a^{m_0} = e$ (it exists by the remark)

Let any other $s \in \mathbb{Z}$ then $\exists! q, r$ $0 \leq r < m_0$

s.t. $a^s = a^{m_0 q + r}$ since $s = m_0 q + r$.

Now $a^{m_0} = e$ then $a^{m_0 q} = e^q = e$.

thus $a^s = a^r$ for $0 \leq r < m_0 - 1$

and $a^r \neq e$ by minimality of m_0 .

Then

$$G = \{e, a, \dots, a^{m_0-1}\}$$

$\uparrow$
 G has as many elements as the smallest m
s.t. $a^m = e$.

Let $f: (\mathbb{Z}/m\mathbb{Z}, + \bmod m, 0) \longrightarrow G$.

$$[r] \bmod m \longmapsto a^r$$

injective: if $a^{r_1} = a^{r_2} \Rightarrow a^{r_1 - r_2} = e$.

where $r_1, r_2 < m_0$ wL6 $r_1 - r_2 > 0$ and

$r_1 - r_2 < m_0$ by minimality of m_0 .

$a^{r_1 - r_2} \neq e$ unless $r_1 = r_2$.

surjective: clear.

hom: $f([r_1] \bmod m + [r_2] \bmod m)$

$$= f([r_1 + r_2] \bmod m) = a^{r_1 + r_2 \bmod m}$$

$$= a^{r_1 \bmod m} \cdot a^{r_2 \bmod m} = f([r_1]) \cdot f([r_2])$$

QED.

what about the subgroups?

- for infinite $\rightarrow$ we are done!

- finite:

[Thm 3] Let G be a cyclic group with n -elements generated by a . Let $b \in G$ and $s \in \mathbb{Z}$ such that $b = a^s$. Then b generates a cyclic subgroup H of G .

containing $\frac{n}{d}$ - elements where $d = \text{HCD}(n, s)$

Also $\langle a^s \rangle = \langle a^t \rangle \iff \text{HCD}(n, s) = \text{HCD}(n, t)$

Proof:

step 1: $H = \langle b \rangle$ has $\frac{n}{d}$ - elements.

By the previous case, if we let m_0 be the smallest integer s.t. $b^{m_0} = e$ then H has m_0 - elements.

Now we know that $b^s = a^s$.

Q: When is it the case that $b^m = e$?
For which m 's?

If $b = a^s$ and $b^m = e \iff (a^s)^m = e$.
thus $a^{sm} = e$.

RMK: since G has n - elements then $a^k = e$ iff n divides k .

SA:

why? $k = q \cdot n + r$ $0 \leq r \leq n-1$

but n is the smallest natural number s.t. $a^n = e$.

then if $a^k = e$ then $a^{qn} \cdot a^r = e \Rightarrow a^r = e$
so $r = 0$.

Then $a^{sm} = e$ iff n divides $s \cdot m$.

Conclusion: $b^m = e \iff a^{sm} = e \iff n$ divides $s \cdot m$.

thus ~~when~~ when we ask which are the m 's
s.t. $b^m = e$? And which is the smallest one.

L_p is the same as asking:

Q2: which is the smallest m such that n divides
 ms ?

Let $d = \text{HCD}(n, s)$ then they are $u, v \in \mathbb{Z}$
s.t. $d = u \cdot n + v \cdot s$

since d divides both n and s

$$1 = u \cdot \left(\frac{n}{d}\right) + v \cdot \left(\frac{s}{d}\right) \Rightarrow \text{both integers, and relatively prime.}$$

when we ask which is the smallest
 m s.t. $s \cdot m$ is divisible by n , this is
the same as asking:

$$\frac{sm}{n} \in \mathbb{Z} \iff \cancel{s} \cdot \cancel{m} \cdot \frac{\left(\frac{s}{d}\right)^m}{\frac{n}{d}} \in \mathbb{Z}$$

Since $\frac{s}{d}$ and $\frac{n}{d}$ are relatively prime if

$\frac{n}{d}$ divides $\left(\frac{s}{d}\right) \cdot m$ by the fundamental

theorem of arithmetic $\frac{n}{d}$ divides m .

then: the smallest m s.t. $b^m = e$

$\Leftrightarrow$ smallest m s.t. n divides sm

$\Leftrightarrow$ smallest m s.t. $\frac{n}{d}$ divides m .

so it is $\frac{n}{d}$. so H has $\frac{n}{d}$ elements.

0

For the second part: we want to show

that $\langle a^s \rangle = \langle a^t \rangle \Leftrightarrow \#CD(n, s) = \#CD(n, t)$

By the previous part we showed that

$H = \langle a^s \rangle$ has $\frac{n}{d}$ -elements which are

exactly $H = \{e, a^s, a^{2s}, \dots, a^{\frac{n-1}{d}s}\}$ $\rightarrow$ cyclic group of $\frac{n}{d}$ -elements.